Ayush Singh

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Education

Chandigarh University Online
Bachelor of Computer Applications *Aug. 2025 – May 2027*

Technical Skills

Red Team & Infrastructure: Active Directory offensive operations including privilege escalation, lateral movement, persistence, attack-path analysis, ADCS abuse, WSUS attacks, ACL/DACL abuse, Kerberos delegation attacks, domain forest trust exploitation, ticket attacks and AV/EDR defense evasion techniques
Offensive Security & Tooling: Burp Suite, Adaptix C2, BloodHound, Impacket, NetExec, Rubeus, Mimikatz
Programming & Scripting: Python, C/C++, PowerShell, Bash, JavaScript, SQL, .NET and Windows API
Development & Research Tool: VS Code, Visual Studio, debugging and reverse-engineering tools, security lab environments, and custom scripting for exploit development, BOFs, shellcode loaders
Technologies Frameworks: MITRE ATTACK, C2 infrastructure concepts, Git/GitHub, REST APIs and web application architecture

Projects

Teamserver Essentials

- Automated one-click C2 Teamserver deployment on GCP using Bash, including free-tier VM provisioning, static IPs, and IAP TCP tunneling.
- Hardened an Apache reverse-proxy redirector with ModSecurity WAF and Let's Encrypt SSL/TLS.
- Implemented secure, firewall-free Teamserver access with automated infrastructure configuration and deployment.

Certification

Offensive Development Practitioner Certification (ODPC) Ongoing
Windows internals and APIs, offensive development, custom payloads, process injection, security evasion, C2 integration, EDR testing, and professional documentation of offensive tooling.

Advanced Red Team Operations Certification (ARTOC) Ongoing
Red team operations, resilient C2 infrastructure, cloud redirector chains, initial access, persistence, lateral movement, Active Directory and ADCS exploitation, EDR/AV evasion, and multi-cloud operations.

Certified Red Team Operations Manager (CRTOM) Achieved
Advanced, scenario-driven red team certification focused on threat-led operations, C2 governance, cloud red teaming, detection validation, Rules of Engagement, safety and deconfliction, security KPIs, compliance, and aligning adversary emulation with business risk.

Achievement

Cybercoliseum IV 2nd Place
Secured 2nd place in the Flag Forge Capture The Flag competition by solving challenges across multiple domains including web exploitation, cryptography, and offensive security techniques.

Hackthebox (Pro Hacker)
Completed many Standalone machine's and Prolabs like RastaLabs—Offshore—Zephyr—Dante—Puppet—P.O.O